PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM TWO PRE -MOCKASSESSMENT
CHEMISTRY

TIME: 2:00 HOURS
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Instructions

1. This paper consists of sections A, B and C with a total of ten (10) questions.
2. Answer all questions in all sections.
3. Section A carries fifteen (15) marks, section B carries seventy (70) marks and section C carries
fifteen (15) marks.
4. All writings must be in blue or black ink except drawings which must be in pencil.
5. All communication devices, calculators and any unauthorized materials are not allowed in the
examination room.
6. Write your Examination Number at the top right corner of every page.
7. Where necessary the following constants may be used.
Atomic masses; H=1, C=12, N=14, O=16, K=39, Cl=35.5, Ca=40, Cu=63.5

SECTION A (15 Marks)
Answer all questions in this section

1. For each of the items (i) – (x), choose the most correct answers from among the given alternatives and write its letter in the box provided.
(i) Makubi went to the laboratory to perform a certain experiment. During experiment she
used a spatula for scooping substances. What is the possible type of substance making
that apparatus scoop?
A. Powdery and gases
B. Crystalline chemicals and powdery
C. Crystalline chemicals and liquids

D. Powdery and liquids.

ii. What is the percentage composition of oxygen in air?

A. 7.8% C. 2.1%

B. 78% D. 21%

iii. A student reacted magnesium with oxygen in order to form magnesium oxide. What happens to magnesium during the formation of magnesium oxide?

A. Magnesium gains protons C. Magnesium loses electrons

B. Magnesium loses protons D. Magnesium gains electrons

iv. Trashes collected from all dustbins in a school were put into a special furnace where they produce energy in the form of light and noticeable heat when lit. What is the process taking place inside the furnace?

A. Decomposition C. Disposing

B. Combustion D. Rotting

v. Wende is facing a problem of fetching water at his village because all available water sources are full of mud. His friend advised him to assemble a simple water filter. Which of the components outlined below should not be included in assembling the device?

A. Cloth C. Charcoal

B. Alum D. Gravel

vi. Class F fire can best be extinguished by using;

A. Carbon dioxide C. Water

B. Sand D. Wet chemical

vii. An element T with atomic number 11 belong to group .....and period .....;

A. II and 2 C. III and 2

B. I and 3 D. III and 3

viii. A form two student was given the following staffs for preparation of oxygen gas;

i) Source of heat (ii)Manganese dioxide

iii) Hydrogen peroxide (iv)Potassium chlorate

Which combination will fast produce oxygen?

A. (ii) and (iv) C. (iii) and (iv)

B. (i), (ii) and (iv) D. (i) and (iii)

ix. Which of the following electronic configurations are non-metals?

A. 2:8:1 and 2:5
B. 2:6 and 2:8:7
C. 2:8:3 and 2:8:8:1
D. 2:8:2 and 2:6

i. Which among the following lists contains the items in transport that are produced as a result of application of chemistry?

A. Dyes, tyres and fuels.
B. Coolants, fuels and lubricants
C. Paints, lubricants and pesticides
D. Lubricants, tyres and drugs

ANSWERS

QUESTION	I	ii	iii	iv	v	vi	vii	viii	ix	X
ANSWERS										

2.Match the methods of preventing rusting in LIST A with the corresponding descriptions in LISTB by writing the letter of the correct response below the item number in the table provided.

LIST A	LIST B
i. Painting ii. Galvanization iii. Sacrificial anode iv. Plating v. Silica gel	A. A more reactive metal than iron is attached to the protected material and will be consumed in Favor of it. B. A substance in form of grains and it absorbs moisture C. The process of coating iron materials or steel with zinc D. The coating of substances such as metals with a special pigment E. Involves coating iron with oil. F. The process of spraying iron with carbon dioxide gas G. The coating of material with tin.

LIST A					
LISTB					

SECTION B (70 Marks)

Answer all questions.

3(a)A teacher planned with his students to perform an experiment at 8:00 pm.Students
reminded the teacher that the laboratory has no source of electricity so how can the experimentbe conducted without light? The teacher told them we shall use the Bunsen burner to get light.

(i) What type of flame will be suitable during their experiment?

_____

(ii) Why is the flame named in (i) above suitable for that purpose? Give two
reasons.

_____

_____

(iii)Draw a well labelled diagram of the flame named in (i) above.

4.(a)Form four (4) students confused the uses of the gas that supports combustion (ignites) with
the one which burns with a Pop sound.As a form two (2) student give the name of the gas that burns with a pop sound, then correct this confusion by giving them four uses of the gas which burns with a pop sound;

Name of the gas

_____

Uses of the gas:

(i) _____

(ii) _____

(iii) _____

(iv) _____

5.Explain briefly why?

i) Laboratory technician advice the user of laboratory apparatus to use evaporating dish instead
conical flask to evaporate solution?

_____

_____

ii) Pipette is the best choice for measuring volume of the base in the titration experiment?

iii) Glassware is so delicate material yet most of the laboratory apparatus made up of glass. Do you think there is any necessity of continuing using glass apparatus? Give two reasons to defend your answer.

b) Is iron a metal or non- metal? Give four (4) reasons.
i)

ii)

iii)

iv) _____

6.Tente wanted to test a substance Q using blue cobalt (II) chloride. When cobalt (II) chloride was exposed to the substance, it changed from blue to pink. Use this information to answer the following questions: -

a) Name the substance that was tested_____

b) Name other substance that could be used in place of cobalt (II) chloride.

c) Outline any three ways by which the substance can be treated and purified.

(i)_____
(ii)_____
(iii)_____

b) State fiveuses of substance Q

(i)_____
(ii)_____
(iii)_____
(iv)_____

7.Oxygen gas can also be prepared by thermal decomposition of potassium chlorate using manganese dioxide as a catalyst.

a) Explain the activities which will be done when preparing the gas using this method.

_____

_____

_____

b) Draw a well labeled diagram to show how oxygen is prepared using this method.

c) Write the word equation for this reaction.

_____

8 a) Briefly give the chemical test for below gases:

i) Oxygen gas _____

ii) Hydrogen gas _____

b) i) Why hydrogen gas collected by downward displacement of water?

ii) Why oxygen gas used in welding and cutting metals?

iii) Give reason why hydrogen gas used to make oxy-hydrogen flame?

_____

9. (a)One self protection is the first principle in the laboratory safety which can be achieved
through the use of soap in washing hand before and after practical, wearing goggles, gloves
and shutting your mouth. Briefly explain how each of the above mentioned acts can help
increase your safety in chemistry laboratory

i)Use of Soap in washing hand before and after experiment

.....

.....

ii).Goggles.....

.....

iii.Gloves.....

.....

iv). Shutting your
mouth.....

.....

(b) Melting of ice is a physical change while souring of milk is a chemical change. Why this
processes are classified as physical and chemical changes give two reasons for each.
Physical change.
1.....
2.....
Chemical change.
1.....
2.....

SECTIONC (15) marks

10(a)Halima's mother was complaining about the type of fuel which could be suitable and
available to use in cooking food for her kids. You as a chemist help her to know four sources of
energy used for cooking in your environments
i).....
ii).....
iii).....
iv).....
(b)classify fuel according to their physical state and in each give two examples

(C)Briefly explain the six characteristics of good fuel.

FUELS……
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iv……
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v……
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vi……
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